```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow # Import cv2_imshow

# Frame properties
width, height = 640, 480
fps = 30
duration = 30          # seconds
total_frames = fps * duration

# Ball position and speed
x, y = 100, 100
dx, dy = 5, 4

for i in range(total_frames):

    # Create black frame
    frame = np.zeros((height, width, 3), dtype=np.uint8)

    # Draw moving ball
    cv2.circle(frame, (x, y), 30, (0, 255, 255), -1)

    # Draw moving rectangle
    cv2.rectangle(frame, (200, 150), (350, 280), (255, 0, 0), 3)

    # Draw moving line
    cv2.line(frame, (0, i % height), (width, (i + 150) % height), (0, 255, 0), 2)

    # Display frame number
    cv2.putText(frame, f"Frame: {i+1}", (20, 40),
                cv2.FONT_HERSHEY_SIMPLEX, 1, (255,255,255), 2)

    # Update ball position
    x += dx
    y += dy

    if x >= width-30 or x <= 30:
```

```
if x >= width-30 or x <= 30:
    dx = -dx
if y >= height-30 or y <= 30:
    dy = -dy

# Source points
src = np.float32([
    [100,100],
    [540,100],
    [100,380],
    [540,380]
])

# Destination points
dst = np.float32([
    [50,50],
    [590,100],
    [120,430],
    [520,380]
])

# Perspective Transformation
M = cv2.getPerspectiveTransform(src, dst)
output = cv2.warpPerspective(frame, M, (width, height))
cv2_imshow(frame)
cv2_imshow(output)
```